

DECSI

Information Technology

Proposal
for Loan Application and Processing System for DECSI

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Proposal for Loan Application and Processing System for DECSI

I. Executive Summary

This updated proposal refines the original plan for the "DECSI Loan Hub" system, incorporating specific requirements to initiate the loan process from queuing for loan requests, ensuring fulfillment of necessary documents, and implementing multi-level authorizations across Branch, Zonal, and Head Office levels. The system will complement the existing Temenos core banking system, streamlining the end-to-end loan lifecycle while enhancing efficiency, compliance, and client experience.

The solution remains a web-based and mobile-friendly application, with added emphasis on queue management, document verification workflows, and hierarchical approvals. Estimated costs at 3-5 million, with an adjusted timeline starting immediately in October 2025, projecting completion in six months. This enhancement will further reduce processing delays, improve risk management, and support DECSI's hierarchical organizational structure.

II. Background and Needs Assessment

Building on the original assessment, DECSI's operations in Ethiopia, particularly in Tigray, involve a structured hierarchy with branches, zonal offices, and a head office in Mekelle. Current challenges include manual queuing at branches, incomplete document submissions leading to rejections, and sequential approvals that cause bottlenecks. The updated system addresses these by:

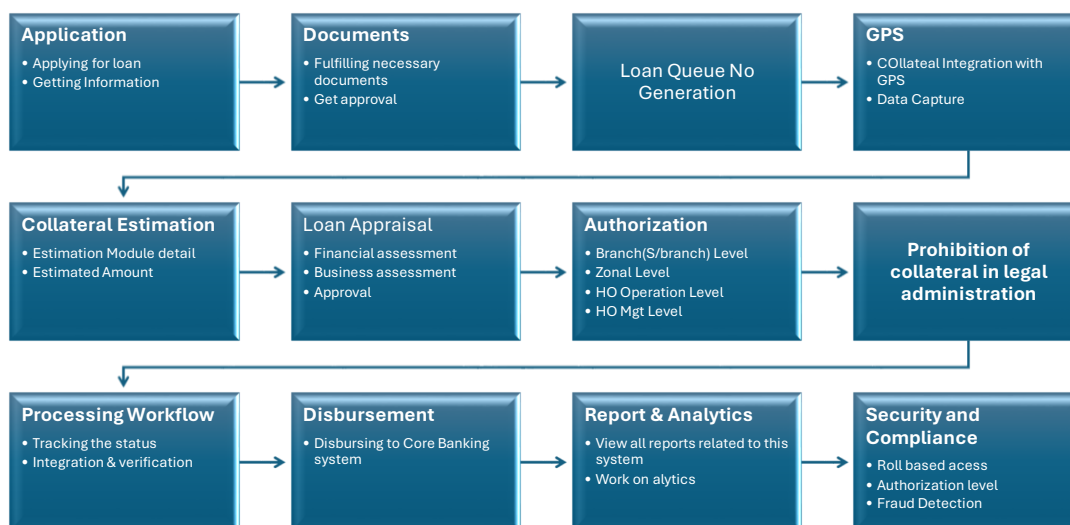
1. Introducing digital queuing to minimize wait times and physical queues.
2. Automating document fulfillment checks to ensure completeness before advancing.

3. Formalizing multi-level authorizations to align with DECSI's Branch → Zonal → Head Office workflow, reducing delays while maintaining oversight.

This aligns with Ethiopian microfinance regulations (e.g., National Bank of Ethiopia Proclamation No. 626/2009) and DECSI's practices, such as group guarantees and collateral requirements for various loan types (e.g., business, housing, agricultural).

III. Proposed Solution

The DECSI Loan Hub will be customized to start the process at queuing, with integrated modules for document handling and tiered approvals. Cloud-hosted for scalability, it will support roles including clients, branch staff, zonal approvers, head office administrators, and loan officers.



Key Features and Functionality

The system covers the full loan lifecycle, starting from queuing, with the following modules:

1. Application to Loan:

- A customer can visit the institution to apply for any loan service and receive detailed information about all available products and the requirements needed to complete the loan application.

- Clients can join a virtual queue for loan consultations, selecting loan types (e.g., regular, business, agricultural) and preferred branches, with automatic addition to the branch's.

2. Fulfilling Necessary Documents:

- Guided document upload portal with checklists tailored to loan types (e.g., business plan, renewed license, ID, marriage certificate, collateral certificates as per DECSI requirements).
- Automated validation using OCR and AI to scan uploads for completeness, authenticity, and compliance (e.g., KYC/AML checks).
- Step-by-step wizard interface to assist clients in fulfilling requirements, with reminders for missing items and secure storage in the system.
- Support for group loan applications, where multiple members upload shared documents (e.g., group guarantees from 3+ members).

3. Queuing for Loan Request:

- Digital queue management system accessible via web portal, or in-branch, with a focus on generating and maintaining branch queue lists.
- Real-time queue status updates via SMS/email, including estimated wait times, appointment scheduling, and queue position visibility for clients.
- Branch staff dashboard for queue list management: View a dynamic list of queued customers, including details like customer name, loan type, arrival time, and pre-eligibility status. Staff can sort, filter (e.g., by loan type or urgency), and manually adjust the list (e.g., prioritize high-potential clients or remove no-shows).
- Integration with Temenos to pre-check client eligibility (e.g., existing accounts, repayment history) during queuing, flagging items on the queue list for fast-tracking or preliminary advisories before starting the process.

- Queue list export functionality for branch reporting, such as daily summaries of queued requests, to help institution management analyze branch traffic and resource needs.

4. Collateral GPS Integration Module

- **GPS Data Capture for Collateral:**
 - Clients or field officers can submit GPS coordinates of collateral assets (e.g., land, property, or equipment) during the document upload phase via the mobile app or web portal.

5. Collateral Estimation Module

- **Automated Valuation Engine:**
 - Integration with GPS data from the Collateral GPS module to automatically estimate collateral value using AI-driven models, market data feeds (e.g., local Ethiopian land/property registries or agricultural commodity prices), and historical DECSI valuation records.
 - Support for various asset types, including real estate (e.g., land plots via geolocation-based comparable sales), livestock/equipment (e.g., depreciation calculators), and group collateral (e.g., aggregated value for guarantees).
 - Manual override option for field officers or appraisers to input expert valuations, with AI suggestions for preliminary estimates to speed up the process.
 - Risk-adjusted scoring: Generates a collateral coverage ratio (e.g., loan amount vs. estimated value) and flags under-collateralized applications for escalation in the approval workflow.
 - Audit trail and reporting: Logs estimation methodology, sources, and any adjustments for compliance with National Bank of Ethiopia guidelines on collateral assessment.

6. Authorization of Different Levels:

The DECSI Loan Hub incorporates a hierarchical workflow engine designed to align with DECSI's organizational structure, ensuring streamlined approvals while maintaining oversight and compliance. The authorization process is automated and routed across four levels: Sub-Branch or Branch, Zonal Office, Operation Head Office, and Management Head Office. Key features include:

Sub-Branch or Branch Level:

- Initial review by loan officers at the Sub-Branch or Branch level, focusing on document verification, preliminary credit assessment, and eligibility checks.
- Officers confirm the completeness of submitted documents (e.g., IDs, collateral certificates, group guarantees) and perform basic risk assessments using integrated credit scoring tools.
- Preliminary approval or rejection at this level, with approved applications escalated to the Zonal Office for further review.
- Dashboard for branch staff to track applications, view queue lists, and manage client interactions.

Zonal Office Level:

- Zonal managers review applications escalated from Sub-Branch or Branch, focusing on higher-risk or larger loans (e.g., above predefined thresholds like loan amount or risk score).
- Additional risk scoring is conducted, leveraging AI-driven models and collateral valuation data from the Collateral Estimation Module.
- Zonal managers can approve, reject, or request additional information, with clear audit trails for decision tracking.
- Configurable escalation rules to ensure only complex cases move to the Operation Head Office.

Operation Head Office Level:

- Operation Head Office handles escalated applications requiring operational oversight, such as high-value loans, exceptions, or cases flagged for compliance issues.
- Reviews incorporate operational metrics (e.g., branch performance, loan portfolio risk) and ensure alignment with DECSI's lending policies and Ethiopian microfinance regulations (e.g., National Bank of Ethiopia Proclamation No. 626/2009).
- Approvals or rejections at this level are logged, with feedback provided to Zonal and Branch levels for transparency.

Management Head Office Level:

- Final authorization for critical cases, such as exceptionally large loans, high-risk profiles, or applications requiring executive sign-off.
- Management Head Office conducts compliance checks, integrates with Temenos for fund allocation, and ensures strategic alignment with DECSI's goals.
- Automated notifications to clients and lower levels upon final decision, with e-signature-enabled loan agreements generated for approved loans.
- Configurable thresholds (e.g., loan amount triggers zonal/head office involvement) and audit trails for each level.
- Notifications and dashboards for approvers at each tier, with escalation timers to prevent delays.

7. Prohibition of collateral in legal administration

The prohibition of collateral in legal administration generally refers to restrictions or limitations on the use of collateral (security for a debt or obligation) in certain legal or administrative contexts.

The customer must process and secure its collateral in accordance with and bring official letter for the prohibition.

8. Processing and Workflow (Integrated with Above):

- End-to-end automation from queue to disbursement, including credit scoring using ML models on DECSI data (e.g., repayment history, group guarantees).
- Real-time status tracking for clients, with SMS/email alerts at each stage.
- Integration with external services if needed.

9. Disbursement:

- Upon final head office approval, seamless fund transfer via Temenos to client accounts or mobile wallets.
- Automated loan agreement generation with e-signatures, compliant with Ethiopian regulations.

10. Reporting and Analytics:

- Dashboards tracking queue volumes, document fulfillment rates, approval times per level, and overall metrics.
- Custom reports for zonal and head office oversight, exportable and integrated with Temenos.

11. Security and Compliance:

- Enhanced with role-based access for each authorization level, data encryption, and logging to meet National Bank of Ethiopia standards.
- Fraud detection alerts during document fulfillment and approvals.

IV. Technical Architecture

- **Frontend:**
- **Backend:**
- **Database:**
- **Deployment:**

V. Implementation Plan

Agile approach with iterations focused on queuing, documents, and approvals.

1. **Discovery Phase (one month):** Refine requirements, map DECSI's hierarchy, and prototype queuing/approval flows.
2. **Development Phase (three months):** Build modules, integrate multi-level workflows.
3. **Testing and QA (one month):** Simulate hierarchical approvals, security testing.
4. **Deployment and Training (one month):** Rollout with training for branch, zonal, and head office staff.

VI. Timeline

Phase	Duration	Start Date (Estimated)	End Date (Estimated)
Discovery			
Development			
Testing & QA			
Deployment & Training			
Total			

Note: Adjusted to start immediately based on current date (_____); assumes quick stakeholder alignment.

VII. Cost Estimate

Unchanged from original proposal:

Component	Estimated Cost ETB
Discovery & Design	
Development	
Integration & Testing	
Deployment & Training	
Cloud Hosting (Year 1)	
Post-Launch Support	
Total	

VIII. Benefits and ROI

- **Streamlined Queuing:** Reduces physical wait times by 50-70%, increasing client satisfaction.
- **Document Efficiency:** Lowers rejection rates due to incomplete submissions by automating checks.
- **Faster Approvals:** Multi-level workflow cuts processing time from weeks to days, with clear accountability.
- **Overall ROI:** Enhanced outreach, reduced defaults, and operational savings projecting break-even in 12-18 months.

IX. Conclusion

This updated DECSI Loan Hub proposal integrates queuing, document fulfillment, and multi-level authorizations to fully align with DECSI's operational structure and country microfinance best practices.